Cover note for review

Draft of 11 November 2026. Working title: *Where the Gains of Self-Critique Come From: Iteration Selects, Critique Reaches, and Same-Model Judges Add Nothing Detectable*

Intended venue: ACL 2027 main conference via ACL Rolling Review (deadline 4 January 2027).

What the paper claims

A self-critique loop (one model answers, criticizes itself, answers again) gains in two separable ways, and placebo controls attribute each to a different source.

1. **Selection comes largely from iteration.** A loop that is only reassured, voted over its own rounds, selects as many correct answers as nine-sample self-consistency at half the tokens. A real critic adds about 12 items on top, which is not significant at 400 items.
2. **Reach comes from critique.** The number of items whose trajectory contains the correct answer rises from 153 (reassurance) to 171 (dose-matched templated suspicions that read nothing) to 198 (generated critique). A vocabulary distilled from the model's past errors adds nothing.
3. **The model cannot convert the extra reach itself.** Last answers are no better than under a placebo, and three same-model judges are at chance wherever they disagree with the vote count.

Status of the evidence

Claim	Status
Ceiling: critique > answer-blind pressure > reassurance	Firm. Each step $p \leq 0.005$; holds at matched candidate counts and under exact-match scoring only.
Free critic selects more than self-consistency	Holds (+20.7, 95% interval [9.0, 32.4]). At least +13 under a conservative bound for vote splitting. A cost-matched comparison (eleven samples) and a universal self-consistency baseline are run and await scoring.
Placebo loop ≥ self-consistency at half the cost	Direction consistent (+8 to +9 items), not significant.
Critic selects more than the placebos	Suggestive only (+11 to +12 items, $p = 0.07$ to 0.08).
Error vocabulary adds nothing	Firm as a null at this sample size.
Same-model judges add nothing to the count	Holds on one model and one condition; gains of a few items cannot be excluded.

Known weaknesses

- One model (deepseek-v4-flash), one benchmark (Humanity's Last Exam, 400 items), one run per condition. A replication on a second model (qwen3.8-flash, same 400 items, pre-specified tests) is running.
- The critic costs about 2.5 times the placebo loop and 1.2 times nine-sample self-consistency.
- The controls do not separate "the critique reads the answer" from "the critique is generated by the model".
- Free-form answers are scored with a model-based tier that has not yet been audited by human annotators. The ceiling result holds without that tier.
- Pre-specification was recorded internally, not registered externally. The original falsification test was written for the bank critic; the clearer result belongs to the free critic, which was introduced as a control. The manuscript says so.
- Section 6 (same-model judges) still uses trajectories recorded before three items were rerun; it will be recomputed.
- No figures yet; the appendix is an outline; references beyond the five checked against sources need verification.

Planned before submission

1. Read out the cost-matched self-consistency and universal self-consistency comparisons (rules fixed in advance).
2. Second-model replication.
3. A confirmatory run on 400 new items with the free critic against the reassuring placebo as the pre-specified primary test, to settle the 12-item selection effect in either direction.
4. Human audit of a stratified sample of scored answers.
5. Conversion to the ACL template (8 pages), figures, full appendix.

Questions on which advice would help most

1. Is the framing "two gains with two sources" the right centre of the paper, or should it lead with the ceiling result alone?
2. Is the evidence sufficient for a main-conference submission if the second model replicates, or is a third model or a second benchmark necessary?
3. Is the confirmatory run on new items worth its cost compared with adding a model?
4. Is the same-model judge study (Section 6) better kept in this paper or separated?

Where the Gains of Self-Critique Come From: Iteration Selects, Critique Reaches, and Same-Model Judges Add Nothing Detectable

Abstract

Loops in which a model answers, criticizes its own answer, and answers again are reported to outperform sampling and voting, although their last answers are known to be unreliable. We separate what such a loop gains into what it reaches and what it selects, and attribute each part with placebo controls. On 400 items from Humanity's Last Exam, with one model as solver and critic, three results emerge. First, selection comes largely from iteration. A loop that is fed a fixed reassuring sentence and votes over its own rounds selects 145 correct answers with half the tokens of nine-sample self-consistency, which selects 137. A critic that must name a specific suspicion and a concrete check raises this to 157, which exceeds self-consistency (difference 20.7, 95% interval [9.0, 32.4]) but exceeds neither placebo significantly (about 12 items, p = 0.07 to 0.08). Second, reach comes from critique. The number of items whose trajectory contains the correct answer rises from 153 under reassurance to 171 under dose-matched templated suspicions that read nothing, and to 198 under generated critique; each step is significant and survives matching the number of candidates and exact-match scoring. A vocabulary distilled from the model's past errors adds nothing identifiable. Third, the model does not convert the additional reach. Last answers improve no more under critique than under a placebo, and three same-model judges of the loop's transitions are at chance wherever they disagree with the vote count (0.46 to 0.50). Critique costs about 2.5 times the placebo loop. Without an external verifier, iteration with a count is the cost-effective component; critique pays where reached answers can be checked.

1 Introduction

A language model that answers a question, receives a critique from itself, and answers again can be evaluated in two ways. Read as a *corrector*, the loop is judged by whether its last answer is better than its first. Read as a *sampler*, it is judged by the set of answers it visits and by what a simple rule selects from that set. The corrector reading has been tested extensively and the result is discouraging: without external information, intrinsic self-correction does not reliably improve reasoning accuracy and often degrades it (Huang et al., 2024; Kamoi et al., 2024). The sampler reading has recently produced positive reports, in which answers refined by self-critique are voted on and beat a majority vote over independent samples (Bilal et al., 2026). These reports leave open where the gain comes from: a loop differs from sampling in that it iterates, in that it puts pressure on the current answer, and in that the pressure has content.

We address this with a decomposition and with controls. Accuracy is the product of how often the correct answer is *reached* and how often it is then *selected*. The first quantity is coverage in the sense of Brown et al. (2024), the fraction of problems solved by any sample; we call its count the reachable ceiling. To attribute changes in either quantity we compare four loops that differ in one respect at a time: a *reassuring placebo* that tells the solver no error was found, a *perturbing placebo* that issues the same number of suspi-

cions as a real critic from fixed templates without reading anything, a *bank critic* whose suspicions must come from a vocabulary of the model's past errors, and a *free critic* that criticizes in its own words. Fixed and adaptive self-consistency (Wang et al., 2023; Aggarwal et al., 2023) serve as sampling baselines. One model plays every role, no reference answers enter the loop, and nothing is trained.

On 400 items from Humanity's Last Exam (Phan et al., 2025) the picture has three parts.

Selection comes largely from iteration. The reassuring placebo, voted over its own rounds, selects 145 correct answers with 25 million tokens; nine-sample self-consistency selects 137 with 51 million. The free critic selects 157, significantly more than self-consistency, at 61 million tokens. Against either placebo its advantage is about 12 items and is not significant at this sample size. Part of the advantage of all loops over sampling may come from independent samples writing the same answer in different forms; under a conservative bound that pools all correct answers of an item, the free critic still exceeds self-consistency by 13 items.

Reach comes from critique. The ceiling rises from 153 under reassurance to 171 under answer-blind pressure and to 198 under generated critique. Each step is significant, and the order is unchanged when the number of candidates is matched and when only exact-match scoring is counted. The vocabulary of past errors makes no measurable difference. The controls do not tell us whether the critique must read the answer or only the question.

The additional reach is not converted by the model itself. The last answer improves no more under critique than under the placebo. An oracle that undoes every harmful switch would recover 19 more items, but three same-model judges with different framings are right half of the time wherever they disagree with the vote count, and aggregating their verdicts never improves on the count out of sample.

Taken together, the results locate the value of self-critique. Iterating and counting is cheap and accounts for most of what a loop selects. Critique buys reach at roughly 2.5 times the cost, and reach becomes accuracy only where candidates can be verified by something other than the model that produced them.

Our contributions are as follows.

1. A controlled attribution of the gains of self-critique loops to iteration, pressure and generated critique, using the coverage–selection decomposition, two placebos and a dose-matched design.
2. Evidence that iteration with a parameter-free count is a cost-effective selector: a placebo loop matches or exceeds self-consistency at half the tokens, and the critic's additional selection gain over placebos is about 12 items and not established at 400 items.
3. Evidence that generated critique raises the reachable ceiling beyond answer-blind pressure of the same dose, robust to candidate count and scoring tier, together with a null result for an error vocabulary distilled from the model's past mistakes.
4. A negative result on same-model evaluation: conditional on the vote count, three judge framings carry no detectable information about which transition was correct.
5. A frozen and audited scoring protocol for free-form answers on Humanity's Last Exam, released with all trajectories, judge verdicts and analysis code.

2 Related Work

Sampling, coverage and voting. Self-consistency (Wang et al., 2023) draws several reasoning paths and returns the most frequent answer. Adaptive variants stop sampling once the vote is settled and recover most of the accuracy at lower cost (Aggarwal et al., 2023); universal self-consistency replaces the vote with a model-based selection for free-form answers (Chen et al., 2023). Brown et al. (2024) separate coverage, the fraction of problems solved by any sample, from the ability to pick the solving sample, and show that coverage keeps growing with the number of samples while majority voting and reward models plateau where no automatic verifier exists. We adopt this decomposition, following the pass@k view of sampling (Chen et al., 2021), and apply it to trajectories produced by revision instead of independent draws.

Self-correction. Self-Refine (Madaan et al., 2023) and Reflexion (Shinn et al., 2023) iterate generation and feedback from the same model. Huang et al. (2024) show that the gains reported for intrinsic self-correction largely vanish when no oracle information leaks into the loop, and Kamoi et al. (2024) survey the conditions under which self-correction helps. Yang et al. (2025) decompose accuracy after one correction step into the rate of keeping correct answers and the rate of fixing incorrect ones; the criterion we use in Section 5.3 is a rearrangement of their identity. Closest to our positive result, Bilal et al. (2026) refine each of eight sampled rollouts with self-critique and vote over the refined answers, reporting gains over majority voting on mathematics benchmarks with small open models at a fixed number of rollouts. Our study differs in what it controls: a single trajectory is voted over its own rounds, the critique is compared with two placebos, the comparison with sampling is reported under token accounting, and the result is separated into what is reached and what is selected.

Verifiers and judges. Process reward models and trained verifiers (Lightman et al., 2023) supply the external signal that self-correction lacks, at the cost of training data with positive labels; we exclude them by design. Language models used as judges are imperfect and prefer their own outputs (Zheng et al., 2023; Panickssery et al., 2024). Zhao et al. (2025) report that sampling many responses and verifying them with the same model scales well for a frontier model, in part because verification improves with the size of the pool and with direct comparison of responses. Our third question concerns a narrower object, verdicts on a model's own answer transitions within one trajectory, and asks what such verdicts add once the vote count is known; Section 6.4 discusses the difference in setting.

3 Method

3.1 Loop and conditions

A solver answers each question greedily (temperature 0). In every subsequent round it receives its previous answer together with a feedback message and answers again; it is told to keep its answer if it believes the answer is correct. A loop stops after three identical consecutive answers or after ten rounds. All conditions share the solver's greedy first answer, so they differ only in what follows it. The solver and the critic are the same model, deepseek-v4-flash, with a reasoning budget of 16,384 tokens and a visible-output margin of 8,192 tokens per call.

We compare the greedy answer with five conditions (Table 1) and one baseline simulated from recorded samples. The *reassuring placebo* returns a fixed sentence stating that no error direction was found. The *free critic* is a separate call to the same model that reads the question, the current answer and the tail of the solver's reasoning, and must return between one and three suspicions, each with a concrete check the solver can carry out without the reference answer; it may not state a probability, name an answer, or remain silent. The *bank critic* is identical except that each suspicion must be drawn from a vocabulary of five error archetypes (Section 3.2). The *perturbing placebo* issues the same number of suspicions per round as the free critic, with archetype names drawn at random and rendered through fixed generic sentences in the critic's exact output format, without reading the question, the answer or the reasoning; it involves no model call. *Self-consistency* draws nine independent samples at temperature 0.7. *Adaptive self-consistency* consumes the same samples in random order and stops when an answer has three votes.

Table 1. Conditions. All loops share the greedy first answer, the stopping rule and the token budget.

Condition	Feedback in rounds ≥ 1	Reads the answer	Model calls per round
Greedy	none	–	1
Reassuring placebo	fixed sentence: no error direction found	no	1
Perturbing placebo	1–3 templated suspicions, archetype names drawn at random	no	1
Bank critic	1–3 answer-specific suspicions with checks, from a five-archetype vocabulary	yes	2
Free critic	1–3 answer-specific suspicions with checks, in the critic's own words	yes	2
Self-consistency	nine independent samples at temperature 0.7	–	9 in total

3.2 Error vocabulary

The bank critic's vocabulary was distilled offline from 33 incorrect single-shot answers of the same model on six benchmarks unrelated to the evaluation set (GPQA-Diamond, MMLU-Pro, MATH-500, BFCL, BBH and AIME). Each error was abstracted into an answer-free description of the faulty operation and a check procedure, and the descriptions were merged across benchmarks into five archetypes: constraint-blind decimation, descriptor or label conflation, procedure truncation or omission, retrieval abort, and format or precision non-conformance. No item, answer or error from Humanity's Last Exam enters the vocabulary. The free critic differs from the bank critic in four textual places only: the vocabulary block, one line of the output format, one clause of the system prompt, and the fallback path (Appendix B).

3.3 Selection rules

We report three rules for loops. *Final* returns the last answer. *Settle* returns the answer of the last run of at least three identical answers and counts as incorrect when no such run exists. *Mode* returns the most frequent answer across rounds. For self-consistency, *vote* returns the most frequent answer among the samples. Answers are compared

through a fixed normalization that depends on the answer type (option letter, integer, or normalized string). A round truncated at the token limit may contribute a vote when its answer can be recovered from the end of its reasoning, but a truncated winner is scored as incorrect.

Ties need care. For a loop, the natural rule is recency, returning the tied answer that occurred last, since later answers have survived more critique; we fixed this rule in the scoring code before any comparison. For exchangeable samples no tie rule carries information, and the two deterministic conventions happen to give 136 and 141 correct on our data. To avoid crediting the loops with a rule the baseline cannot use, all headline comparisons use a *symmetric convention*: ties are broken uniformly at random in every condition, and each item receives its expected score, which is computed exactly and can be fractional. We report the recency figures alongside and quantify what recency contributes.

4 Experimental Setup

Data. We use 400 items from Humanity's Last Exam, of which 84 are multiple choice and 316 are open-form. Trajectories in which any call failed at the interface were discarded and rerun in full, in every condition (29 items for the perturbing placebo and between one and four for the others); an error message never counts as an answer.

Scoring. Every answer of every round is scored once, by a protocol frozen before the conditions were compared. Answers are first matched exactly according to answer type. Unmatched answers are then examined by a model-based rescue of false negatives and by an independent judge from a different model family; disagreements are re-judged three times, and residual disputes are adjudicated by answer variant under the rule that the same substance receives the same verdict in every condition and round (Appendix C). Because the rescue step can only turn rejections into acceptances, scoring errors could inflate the ceiling of conditions that produce more distinct candidates; Section 8 returns to this.

Statistics. Conditions are compared on the same 400 items. Ceilings and last answers are binary and are compared with exact McNemar tests. Selected answers are compared through the per-item expected scores with a paired sign-flip permutation test (10,000 permutations), and differences carry 95% bootstrap intervals over items. Ratios carry bootstrap intervals over items. P-values are unadjusted. Anything fitted is fitted on items with even index and evaluated on items with odd index.

Analysis status. The falsification test was fixed for the bank critic, which was the method under study at the time; the free critic and the perturbing placebo were introduced afterwards as controls, each with a read-out rule written before its data were seen. The falsification test, the reassuring placebo, the read-out rules for the free critic and the perturbing placebo, and the acceptance criteria for evaluators were written down, with dated records, before the corresponding data were seen. The generation–selection decomposition as applied here, the one-step criterion, the conditional-on-count test, selective prediction and the verifier simulation are exploratory. The symmetric tie convention, the adaptive baseline, the accounting of the recency rule, the matched-candidate and exact-match ceilings, the merging of equivalent answers and the robustness analyses of Section 6 were added after review; each tests the conclusions against an alternative ex-

planation. Three selection rules were listed in advance without a designated primary rule; we report all three.

5 Results

5.1 Reaching and selecting

Table 2. Correct items out of 400. *Ceiling* is the number of items for which some candidate in the trajectory is correct. *Selected* uses mode for loops and vote for sampling and is the expected score under random ties, with the recency rule in parentheses. Tokens are in millions; critic tokens are extrapolated from replays of 50 calls per critic.

Condition	Final	Settle	Selected	Ceiling	Solver tokens	Critic tokens
Greedy	113	–	113	113	5.1	0
Reassuring placebo	141	135	145.0 (144)	153	24.6	0
Perturbing placebo	142	134	146.1 (147)	171	31.8	0
Self-consistency	–	–	136.7	190	51.4	0
Adaptive self-consistency	–	–	135.8	–	35.5	0
Bank critic	140	117	150.9 (154)	192	37.1	≈ 29
Free critic	148	121	**157.4 (159)**	**198**	38.1	≈ 23

The ceiling. Both critics reach the correct answer far more often than the reassuring placebo (192 and 198 against 153; discordant items 47/8 and 52/7, $p = 8 \times 10^{-8}$ and 1×10^{-9}) and as often as nine-sample self-consistency (190; 24/22 for the bank critic and 29/21 for the free critic, $p = 0.88$ and 0.32). The two critics do not differ from each other (23/17, $p = 0.43$).

A ceiling is a maximum over candidates, and the conditions differ in how many distinct candidates they produce (Section 5.4), so we repeat the comparison with the pool size held fixed and with the model-based scoring tier removed (Table 3). Taking only the first k distinct candidates of every trajectory, the free critic stays ahead of the perturbing placebo at $k = 2$, 3 and 4 (33/9, 35/11 and 37/10; $p \le 0.0005$) and ahead of the reassuring placebo by a wider margin. Counting only answers accepted by exact matching lowers every ceiling and preserves the order (free critic against perturbing placebo 31/11, $p = 0.003$; against reassuring placebo 38/7, $p < 10^{-5}$). The advantage arises at the first revision. When the first candidate is wrong, the critics produce a second candidate more often, and that second candidate is correct on 48 of 204 items for the free critic (0.24) and 43 of 198 for the bank critic (0.22), against 26 of 171 for the perturbing placebo (0.15) and 23 of 135 for the reassuring placebo (0.17). Later candidates are correct at similar and low rates in all conditions.

Table 3. Robustness of the ceiling: first k distinct candidates per trajectory, and exact-match scoring only.

	Reassuring placebo	Perturbing placebo	Self-consistency	Bank critic	Free critic
k = 2	141	148	169	166	172
k = 3	150	157	178	177	181
k = 4	151	164	183	180	191
all candidates	153	171	190	192	198
all candidates, exact match only	91	102	120	115	122

The selected answer. All four loops select at least as many correct answers as self-consistency, and the cheapest of them does so with half the tokens: the reassuring placebo selects 145.0 against 136.7 for nine samples and 135.8 for the adaptive variant (9.1 [−3.5, 21.5], p = 0.14, against the latter), a difference in the loop's favour that is not significant. The free critic selects 157.4 correct answers against 136.7 for self-consistency (difference 20.7, 95% interval [9.0, 32.4], p < 0.001) and 135.8 for adaptive self-consistency (21.5 [10.0, 33.6], p < 0.001). The bank critic, for which the falsification test was written, selects 150.9 (14.1 [2.5, 26.7], p = 0.026). Neither placebo differs from self-consistency (reassuring 8.2 [−4.0, 21.1], p = 0.20; perturbing 9.4 [−2.4, 20.7], p = 0.11). The recency rule contributes +3.1 items to the bank critic, +1.6 to the free critic and −1.0 to the reassuring placebo; the advantage of the critics lies in the counts.

Vote splitting. The gain over self-consistency is confined to open-form items; on multiple-choice items all conditions are within three items of one another. Open-form answers can be written in several ways, and independent samples are more exposed to this than a loop in which the solver sees its previous answer: 24% of correct samples in self-consistency fall outside the item's most common correct form, against 20% of correct rounds under the critics and 11% under the reassuring placebo. Two analyses bracket the effect. The first is reference-free: stripping display wrappers, boxing and spacing macros from open-form answers before voting, for correct and incorrect answers alike, merges few candidates and leaves the comparison unchanged (free critic 159.1 against 137.5 for self-consistency; 21.6 [10.0, 32.6], p < 0.001). An earlier canonicalization of purely numeric answers merged candidates on four items. Formatting therefore explains little, and what remains are re-expressions that only a semantic judgement can equate. The second analysis is a conservative bound: we pool all answers scored correct for an item into one candidate before voting, in every condition (Table 4). This uses the reference answers, leaves incorrect answers split, and also pools answers that differ in substance and were each accepted by the scorer, so it overstates splitting and is a diagnostic, not a selection rule. Under this bound self-consistency gains 13 items and the free critic 5.5; the difference between them falls to 13.0 [2.1, 23.8] (p = 0.019), and 14.7 [3.9, 25.1] against the adaptive variant (p = 0.007). For the bank critic the difference falls to 10.0 [−0.9, 20.9] (p = 0.08). The pre-specified test therefore passes nominally and does not survive the bound, and the stronger evidence comes from the free critic, which was introduced as a control. Under the bound, both critics exceed the reassuring placebo (free critic 17.4 [3.6, 30.3], p = 0.011; bank critic 14.4 [1.7, 26.4], p = 0.019), which they do not without it (12.4 [−1.6, 25.6], p = 0.08, and 5.9, p = 0.34): critique scatters correct answers over more forms than reassurance does, and the plain count understates its effect.

Table 4. Selected answer under the plain vote and under the bound that pools all answers scored correct for an item into one candidate.

	Reassuring placebo	Perturbing placebo	Self-consistency	Adaptive self-consistency	Bank critic	Free critic
Plain vote	145.0	146.1	136.7	135.8	150.9	157.4
Correct answers pooled	145.5	149.1	149.8	148.1	159.9	162.9

The other rules. The last answer is worse than the majority in every loop (141 against 145, 142 against 146, 140 against 151, 148 against 157), and no loop's last answer differs significantly from self-consistency (free critic 11.3 [−2.1, 24.2], p = 0.10). Settle, which counts an item as incorrect when its trajectory contains no run of three identical answers, falls below self-consistency for both critics (117 and 121).

Cost. The critics use about as many solver tokens as adaptive self-consistency (37–38 million against 36 million) and in addition pay for one critic call per round, which the replays put at about 23 million tokens for the free critic and 29 million for the bank critic, most of it hidden reasoning. Their totals are about 1.7 and 1.9 times that of adaptive self-consistency, 1.2 and 1.3 times that of nine-sample self-consistency, and 2.5 and 2.7 times that of the reassuring placebo. We did not run self-consistency with the eleven samples that would match the free critic's total.

5.2 What in the critique matters

The vocabulary. The free critic is at least as good as the bank critic on the ceiling and on every selection rule, and the differences are not significant (Table 5). The read-out rule for this comparison classified the vocabulary as "not identifiable" if the free critic's ceiling was no more than five items below the bank critic's and their persistence intervals overlapped; both conditions hold. The two critics also differ in dose: freed from the vocabulary, the free critic issued three suspicions in 86% of rounds, whereas the bank critic issued one in 57% and three in 12%. This is a further reason not to read the free critic's nominal lead as superiority. The perturbing placebo is matched to the free critic's dose, so the comparison below is not affected.

Pressure against generated critique. The perturbing placebo applies the same pressure, in the same format and at the same dose as the free critic, without reading the question, the answer or the reasoning. It raises the ceiling over the reassuring placebo from 153 to 171 (p = 0.005), which shows that pressure alone makes the solver explore. It stops well short of the free critic's 198 (39/12, p = 0.0002), and the gap persists at matched candidate counts and under exact-match scoring (Table 3). The read-out rule for this control classified content as effective if the perturbing placebo's ceiling fell at least 15 items below the free critic's at p < 0.05; it fell 27 below. Of the 45-item ceiling gain of the free critic over the reassuring placebo, 18 items are attributable to answer-blind pressure and 27 to generated critique.

On the selected answer the evidence is weaker. The perturbing placebo does not differ from the reassuring placebo (1.2 [−10.1, 12.5], p = 0.84). The free critic selects 11.3 more correct answers than the perturbing placebo, with an interval that includes zero ([−0.8, 23.7], p = 0.07); under the pooling bound of Section 5.1 the difference is 13.8

[2.1, 25.0] (p = 0.016). The extra candidates that pressure produces do not become votes for the correct answer, and whether generated critique converts more of them is suggested and not established.

What the control does not isolate. The free critic differs from the perturbing placebo in three respects: it reads the question, it reads the answer and the end of the solver's reasoning, and it is generated by the model with a reasoning budget of its own. A critic with that budget may partly re-solve the problem and pass its solution to the solver through the check it proposes, in which case it would act as a second solver more than as a reader of the first. Two observations on the 301 switches of the bank critic that change correctness speak against this account without excluding it. The critic's text contains the solver's new answer in none of the 129 repairing switches with a numeric or free-text answer and in 2 of the 112 damaging ones, so an answer is not passed on literally; what the solver receives is a check to carry out. And repairing and damaging switches are almost equally frequent (159 and 142), whereas a critic that had solved the problem would be expected to repair far more often than it damages. A critic that reads the question and is denied the answer would separate the accounts directly, and we have not run it. We therefore attribute the effect to generated critique of this question and this answer, and leave open which of the two inputs carries it.

Table 5. Paired comparisons for the controls. Ceiling: totals, discordant items in parentheses, exact McNemar. Selected answer: difference in expected score, 95% interval, permutation test.

Comparison	Ceiling	Selected answer
Free critic vs bank critic	198 vs 192 (23/17), p = 0.43	+6.5 [−6.4, 18.9], p = 0.32
Free critic vs perturbing placebo	198 vs 171 (39/12), p = 0.0002	+11.3 [−0.8, 23.7], p = 0.07
Perturbing vs reassuring placebo	171 vs 153 (28/10), p = 0.005	+1.2 [−10.1, 12.5], p = 0.84
Free critic vs reassuring placebo	198 vs 153 (52/7), p = 1×10^{-9}	+12.4 [−1.6, 25.6], p = 0.08
Bank critic vs reassuring placebo	192 vs 153 (47/8), p = 8×10^{-8}	+5.9 [−6.8, 18.5], p = 0.34

5.3 The last answer and the self-correction literature

Let the repair rate be the probability that an item wrong at the greedy answer is correct after one revision, and the damage rate the probability that a correct item becomes wrong. One revision improves accuracy only if the ratio of repair to damage exceeds a/(1 − a), where a is the initial accuracy; this is the identity of Yang et al. (2025) solved for the break-even point. Here a = 0.283 and the threshold is 0.39. The first revision repairs 17 items and damages 7 under the reassuring placebo (ratio 0.96), 38 and 19 under the bank critic (0.79), and 45 and 20 under the free critic (0.89). All are above the threshold, and the critics move two to three times as many items in each direction as the placebo. On a benchmark where the model starts at 80%, the same ratios would have to exceed 4, and a critique that moves items in both directions would lower accuracy. This is the regime in which the negative results on self-correction were obtained.

Two cautions apply. Rates pooled over all rounds are not a substitute for this one-step check, because items that settle early contribute few rounds and items that keep cycling contribute many. And the criterion does not show that critique corrects: the last answer under the critics does not differ from the last answer under the reassuring placebo (140

and 148 against 141; 27/28 and 29/22, p = 1.0 and 0.40). At low initial accuracy any revision with a nearly balanced profile helps a little, and the effect of the critique appears in the trajectory, in what is reached and how often it recurs.

5.4 Mechanism: activation, persistence and stopping

Activation. The critics widen the set of answers the solver visits: 3.5 and 3.7 distinct candidates per item, against 2.1 for the reassuring placebo and 2.7 for the perturbing placebo. The cumulative ceiling of the bank critic exceeds the placebo's final ceiling after two rounds. Self-consistency reaches the same ceiling with more candidates, which shows that non-specific variation suffices to reach an answer; the perturbing placebo shows that the solver responds to answer-blind suspicions far less than to suspicions that read its answer.

Persistence. The selector exploits the probability that the next answer repeats the current one, which differs according to whether the current answer is correct (Table 6). Correct answers are more persistent than incorrect ones in every condition. The reassuring placebo makes all answers persistent, so its count carries little information per round. The bank critic's asymmetry equals that of independent sampling, and the free critic's is higher, with overlapping intervals. What distinguishes the critics from sampling is therefore the sequential structure on which the stopping rule acts: repeats are consecutive, three in a row end the loop, and the trajectory has a direction. Stopping alone does not explain the gap, since adaptive self-consistency, which stops exchangeable samples once an answer has three votes, gains nothing over fixed sampling.

Table 6. Probability that the next answer repeats the current one, by correctness of the current answer, among rounds with a committed answer (for self-consistency, over ordered pairs of samples); 95% bootstrap intervals over items.

Condition	Current correct	Current incorrect	Ratio
Self-consistency	0.50 [0.44, 0.56]	0.28 [0.24, 0.32]	1.8 [1.5, 2.1]
Bank critic	0.52 [0.46, 0.58]	0.30 [0.27, 0.33]	1.7 [1.5, 2.0]
Free critic	0.53 [0.48, 0.60]	0.25 [0.22, 0.28]	2.1 [1.8, 2.5]
Perturbing placebo	0.69 [0.64, 0.74]	0.41 [0.38, 0.45]	1.7 [1.5, 1.9]
Reassuring placebo	0.77 [0.71, 0.83]	0.54 [0.50, 0.59]	1.4 [1.3, 1.6]

Normalization. In the critic conditions 27% of transitions between two correct answers change the normalized form of the answer (16% under the placebo). Equivalent correct answers written differently split the vote, so the count understates the support for correct answers. Merging them would require an equivalence judge, which is the problem of Section 6 in another form.

5.5 The vote margin as an abstention signal

If items are ranked by the margin between the two most frequent answers and only the top fraction is answered, the critics yield higher accuracy than both baselines at every coverage (Table 7). The curves are not monotone at low coverage: the free critic falls from 0.510 at one half to 0.492 at one third, where the bank critic is ahead. The area under the ROC curve of the margin as a predictor of a correct selection is 0.64 for both crit-

ics and for self-consistency and 0.59 for the reassuring placebo, with overlapping intervals. Adding judge verdicts to the margin does not improve it out of sample.

Table 7. Accuracy when only the most confident fraction of items is answered (ties by recency for loops).

Coverage	Reassuring placebo	Self-consistency	Bank critic	Free critic
1.00	0.360	0.342	0.385	0.398
0.80	0.391	0.387	0.425	0.453
0.50	0.425	0.445	0.485	0.510
0.33	0.386	0.470	0.515	0.492

5.6 From ceiling to accuracy under external verification

The ceiling becomes accuracy when candidates can be verified. To see how the conditions would fare with an imperfect verifier, we simulate one on the recorded trajectories: it accepts each candidate independently with accuracy q, the most frequent accepted candidate is returned, and the majority is returned when nothing is accepted (Table 8). With independent errors, the critics' advantage over the reassuring placebo grows with q, from 4 items at q = 0.5 to 31 at q = 0.9 for the bank critic, and their advantage over self-consistency shrinks towards the difference in ceilings without changing sign. When the verifier is assumed weaker on items with more candidates and on items the greedy answer got wrong, the advantage over self-consistency no longer shrinks, because sampling carries the most candidates and suffers most from false acceptances. These are simulations under stated assumptions, and we draw one qualitative conclusion: trajectories produced under critique are at least as good a candidate pool as samples, and the less reliable the verification, the larger their advantage.

Table 8. Simulated correct items when a verifier of per-candidate accuracy q filters the candidates.

Verifier errors	q	Reassuring placebo	Self-consistency	Bank critic	Free critic
independent	0.5	135	126	139	145
independent	0.7	143	152	161	166
independent	0.9	150	177	181	187
correlated with difficulty	0.7	140	135	147	154
correlated with difficulty	0.9	146	153	164	169

6 Can a Same-Model Evaluator Improve on the Count?

This section asks whether the model itself can recover any of the items the bank critic reaches and its majority does not select, using the bank critic's trajectories throughout. The analysis was run on the trajectories as first recorded; three of them were later rerun after interface failures (Section 4), so the counts here are 191 reached and 154 selected, one item below Table 2.

6.1 Headroom

Of the 37 recoverable items, 11 are ties, 10 have a margin of one vote and 16 a larger margin. The trajectories contain 1,201 switches between two committed answers, of which 159 turn an incorrect answer into a correct one and 142 do the opposite. An oracle that undoes every harmful switch would reach 173 correct items, and one restricted to switches the solver explicitly attributes to the feedback would reach 165. Undoing all feedback-attributed switches without knowing which are harmful gives 149, a loss of five. Recovery therefore requires telling a warranted suspicion from an unwarranted one.

6.2 Judges, aggregation and acceptance criteria

An evaluator is a source of verdicts on switches, combined with the vote by a rule. We test three judge framings, all with the same model and without reference answers. *Judge-R* scores the new answer on three dimensions (disconnection from established steps, contradiction, and a locatable error) and votes to cut it when any score is high. *Judge-R+* additionally sees the question, must quote the step it objects to, and must name a class of falsifier. *Judge-C* judges the critique instead of the answer: given the question, the suspicion and the criticized answer, it decides whether the suspicion was warranted. Verdicts require agreement across two presentations in swapped order.

The aggregation rule adds to the logarithm of each candidate's count the log-likelihood ratios of the verdicts on the switches entering and leaving it, where a ratio compares the probability of a verdict on repairing and on damaging switches. Ratios are estimated on development items and frozen for test items; when all ratios equal one, the rule reduces to the majority. Before running the judges we fixed four criteria: accuracy on decisive switches, intervals of the likelihood ratios, the test-set comparison with the majority, and stability of the ratios across halves.

6.3 Results

Table 9. Same-model judges on the bank critic's trajectories. Decisive switches are those that change correctness. The majority selects 76 correct items on the test half and 78 on the development half.

	Judge-R	**Judge-R+**	**Judge-C**
Switches judged (of 1,201)	1,185	1,041	1,201
Accuracy on decisive switches	0.58	0.60	0.66
Likelihood-ratio intervals exclude 1	no	yes	yes
Fitted on development, applied to test	76 (+2, −2)	73 (+3, −6)	71 (+3, −8)
Fitted on test, applied to development	73 (+5, −10)	72 (+5, −11)	77 (+8, −9)
Accuracy where the judge disagrees with the count	18/36 = 0.50 [0.35, 0.65]	53/116 = 0.46 [0.37, 0.55]	32/66 = 0.49 [0.37, 0.60]
Best of 81 aggregation rules, in-sample	154	156	155

No judge improves on the majority. Judging the critique is easier than judging the answer, and two of the three judges carry statistically detectable information at the level of single switches. That information does not survive aggregation in either direction of the split, and the reason is visible in the sixth row. A rule that uses only the counts, predicting that a switch is a repair when the new answer ends with at least as many votes as the old one, is as accurate on decisive switches as the judges are. Where a judge disagrees with this

rule, which is the only place it can change the selection, its accuracy is 0.50, 0.46 and 0.49, and every interval contains one half. The aggregation rule fits the direction of each verdict from data, so a judge that was reliably wrong would have been exploited as readily as one that was reliably right. The verdicts restate the count.

Two further checks support this reading. A logistic model over count features and all verdict features of Judge-C, fitted and evaluated on all 400 items, selects 150 correct answers, fewer than the majority. A sweep over 81 aggregation rules, varying where a verdict acts (on the boundary round, the whole run, or the whole candidate), the weights of the two verdict directions, and gating to low-margin items, gains at most two items in-sample and nothing when the best rule on one half is applied to the other.

Why an accuracy of 0.66 on single switches is insufficient follows from base rates. Overturning the majority is a repair on 37 of the 123 items with several candidates where it can be overturned. A judge with accuracy p on a switch therefore overturns correctly with precision $0.30p / (0.30p + 0.70(1 - p))$, which is 45% at $p = 0.66$ and exceeds 50% only for p above 0.70. On the 56 tied items the count is silent and Judge-C is right on 17 of 26 disagreements, which is suggestive and not significant ($p = 0.17$), and this changes the selection on few items, and the recency rule alone already recovers 14 of the 23 ties whose tied set contains the correct answer.

6.4 Interpretation

Conditional on the vote count, none of the three same-model judges carried detectable information about which answer was correct. This is a statement about one model, one benchmark, three framings and the trajectories of one condition, with the power of 400 items; gains of a few items per judge are not excluded, and a judge that outperforms the count on these switches would refute it. It does not contradict the finding of Zhao et al. (2025) that same-model verification scales with a large pool of independent responses: our judges see one trajectory of three to four candidates produced by a solver that has already reacted to critique from the same model, and they are asked about transitions, not about responses side by side. The three framings are not independent evidence, since they share a model and each correlates with the count, which is the finding. The headroom is real, and recovering it requires information that is not a function of the trajectory the solver has already produced: a stronger model, external verification, or a critic that withholds unwarranted suspicions in the first place.

7 Discussion

Two gains with two sources. The loop's selection gain and its reach gain respond to different interventions. Selection responds to iteration: every loop, including one that is only reassured, selects at least as well as sampling, and the count over rounds works because correct answers recur more than incorrect ones in every condition (Table 6). Reach responds to critique: it grows with pressure and grows further when the pressure is generated for the question and the answer at hand. The two do not move together. The perturbing placebo adds 18 reached items and no selected ones; the free critic adds 45 reached items and about 12 selected ones, the latter not established. Extra candidates enter the count as extra competitors, and without a signal that separates them the count returns most of what was gained.

Why the model cannot collect the difference. The negative result on evaluation is the counterpart of this. A judge from the same model sees the same evidence that made the solver return to an answer, so its verdicts are informative in isolation and redundant given the count. Progress on selection in this setting should be sought in information the trajectory does not contain: a stronger model, an executable check, or retrieval.

Reconciling the literature. The corrector reading and the sampler reading are both right about what they measure. Last answers do not improve beyond a placebo, as the negative literature holds. Voting over refined answers does beat voting over independent samples, as recent positive reports hold, and our controls suggest that much of that margin would also be obtained with a placebo in place of the critic. Whether a study finds self-critique helpful therefore depends on whether it compares against sampling or against a loop without content, and on whether it scores the last answer or the trajectory.

Practical reading. With no verifier, the reassuring loop with a count is the efficient choice: it costs half of nine-sample self-consistency and selects no worse. A critic is worth its 2.5-fold cost when reached answers can be verified, since it reaches 45 more items than the placebo loop and as many as nine independent samples with fewer distinct candidates, and the simulation of Section 5.6 indicates that its advantage is retained under imperfect verification.

8 Limitations

The study uses one model, one benchmark and one run per condition, and none of the roles was tuned; the results establish that the effects exist under these conditions and do not establish that they are general. The benchmark's reference answers are known to contain errors, which adds noise to every condition.

The critique's most secure effect is on the ceiling. On the selected answer its advantage over self-consistency lies between 13 and 22 items depending on how equivalent answers are treated; the lower figure comes from a bound that relies on reference answers, and a reference-free semantic merge with a universal self-consistency baseline remains to be done. Against either placebo the selected answer is not significantly better without pooling. The falsification test was written for the bank critic and passes only nominally; the clearer result belongs to a control arm.

The perturbing placebo controls for pressure, dose and format. It does not control for generation, for reading the question, or for the length and fluency of the feedback, so the effect is attributed to generated critique and not specifically to reading the answer.

The critics cost more than either sampling baseline, and we did not run a sampling baseline matched in total tokens. Critic token counts are extrapolated from a replay.

Scoring free-form answers required a model-based tier that can only add acceptances. The ceiling comparison holds under exact matching alone, but the tier has not been audited by human annotators. The free critic and the perturbing placebo were run through a different service endpoint for the same model weights than the other conditions; per-round reasoning length, truncation rate and commit rate agree across endpoints.

The evaluator analysis uses the bank critic's trajectories only and judges from the solver's own model; it says nothing about stronger judges. Pre-specification was recorded inter-

nally with dated documents and was not registered externally, and p-values are not adjusted for the number of comparisons. Finally, 400 items resolve paired differences of roughly twelve items or more in the selected answer; smaller effects, including small gains from same-model judges, remain undetermined.

9 Conclusion

We asked where the gains of a self-critique loop come from and found two sources that should not be conflated. What the loop selects comes largely from iterating and counting, which a placebo provides at half the cost of self-consistency. What the loop reaches comes from critique: 45 more items than reassurance and 27 more than answer-blind pressure of the same dose, at matched candidate counts and under exact-match scoring alike, with no identifiable contribution from a vocabulary of the model's past errors. The model does not convert that reach on its own: its last answer is no better than a placebo's, and its judgements of its own transitions add no detectable information to how often it returned to an answer, within the resolution of 400 items. Whether critique also improves selection by the roughly 12 items our data suggest, and whether it must read the answer to do so, are open and testable. Where answers can be verified, the answers that are reached but not selected are the gain to be collected; where they cannot, a count over a cheap loop serves as both selector and abstention signal.

References

Aggarwal, P., Madaan, A., Yang, Y., and Mausam (2023). Let's sample step by step: Adaptive-consistency for efficient reasoning and coding with LLMs. *EMNLP*.

Bilal, A., Mohsin, M. A., Umer, M., Trigg, L., Subhan, A., Ali, M., and Hougen, D. F. (2026). Refining over resampling: Test-time self-correction for LLM reasoning. *arXiv:2608.05643*.

Brown, B., Juravsky, J., Ehrlich, R., Clark, R., Le, Q. V., Ré, C., and Mirhoseini, A. (2024). Large language monkeys: Scaling inference compute with repeated sampling. *arXiv:2407.21787*.

Chen, M., Tworek, J., Jun, H., et al. (2021). Evaluating large language models trained on code. *arXiv:2107.03374*.

Chen, X., Aksitov, R., Alon, U., et al. (2023). Universal self-consistency for large language model generation. *arXiv:2311.17311*.

Huang, J., Chen, X., Mishra, S., et al. (2024). Large language models cannot self-correct reasoning yet. *ICLR*.

Kamoi, R., Zhang, Y., Zhang, N., Han, J., and Zhang, R. (2024). When can LLMs actually correct their own mistakes? A critical survey of self-correction of LLMs. *TACL*.

Lightman, H., Kosaraju, V., Burda, Y., et al. (2023). Let's verify step by step. *arXiv:2305.20050*.

Madaan, A., Tandon, N., Gupta, P., et al. (2023). Self-Refine: Iterative refinement with self-feedback. *NeurIPS*.

Panickssery, A., Bowman, S. R., and Feng, S. (2024). LLM evaluators recognize and favor their own generations. *NeurIPS*.

Phan, L., Gatti, A., Han, Z., et al. (2025). Humanity's Last Exam. *arXiv:2501.14249*.

Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K., and Yao, S. (2023). Reflexion: Language agents with verbal reinforcement learning. *NeurIPS*.

Wang, X., Wei, J., Schuurmans, D., et al. (2023). Self-consistency improves chain of thought reasoning in language models. *ICLR*.

Yang, Z., Zhang, Y., Wang, Y., Xu, Z., Lin, J., and Sui, Z. (2025). Confidence v.s. critique: A decomposition of self-correction capability for LLMs. *arXiv:2412.19513*.

Zhao, E., Awasthi, P., and Gollapudi, S. (2025). Sample, scrutinize and scale: Effective inference-time search by scaling verification. *arXiv:2502.01839*.

Zheng, L., Chiang, W.-L., Sheng, Y., et al. (2023). Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *NeurIPS Datasets and Benchmarks*.

Appendix

A. Error vocabulary. Source items by benchmark, the three distillation levels, the five archetype entries with their definitions and check procedures, and the audit confirming that no entry contains a reference answer or an item identifier.

B. Prompts and runtime. Solver, critic and judge prompts; the four textual differences between the bank critic and the free critic; the generic sentences of the perturbing placebo; token budgets and stopping constants; service endpoints and their alignment statistics.

C. Scoring protocol. The three tiers, the truncation rule, the answer normalization and tie rules, and the table of adjudicated answer variants with rationales.

D. Evaluator details. Per-judge confusion tables on development, test and all items; likelihood ratios with intervals; a three-role Bayesian aggregator; the sweep of aggregation rules; a contrast of 26 gold-free trajectory features between repairing and damaging switches (areas under the curve between 0.43 and 0.59, none stable across halves); and a probe showing that critic suspicions concern the derivation and not the question text.

E. Pre-specified rules. The falsification test, the read-out rules for the free critic and the perturbing placebo, and the evaluator acceptance criteria, with dates.

F. Additional results. Results by answer form for every condition and rule; ceiling by round; selective-prediction curves; verifier simulations with asymmetric error rates.